

Equation (8.2.1) exactly fits the form of the eigenvalue problem given by Eq. (8.1.28). Thus, the **generalized eigenproblem for longitudinal motion** is

$$\begin{aligned} & \begin{bmatrix} R_{x,\mu} & R_{x,\alpha} & R_{x,\bar{q}} & 0 & 0 & -R_{gx} \cos \theta_o \\ R_{z,\mu} & R_{z,\alpha} & (1 + R_{z,\bar{q}}) & 0 & 0 & -R_{gx} \sin \theta_o \\ R_{m,\mu} & R_{m,\alpha} & R_{m,\bar{q}} & 0 & 0 & 0 \\ \cos \theta_o & \sin \theta_o & 0 & 0 & 0 & -\sin \theta_o \\ -\sin \theta_o & \cos \theta_o & 0 & 0 & 0 & -\cos \theta_o \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \\ & -\lambda \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & (1 - R_{z,\hat{\alpha}}) & 0 & 0 & 0 & 0 \\ 0 & -R_{m,\hat{\alpha}} & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} \chi_\mu \\ \chi_\alpha \\ \chi_{\bar{q}} \\ \chi_{\xi_x} \\ \chi_{\xi_z} \\ \chi_\theta \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{Bmatrix} \end{aligned} \quad (8.2.2)$$

The eigenvalues and eigenvectors associated with longitudinal motion can be found by solving Eq. (8.2.2) using a generalized eigensolver. If a generalized eigensolver is not available, this system of equations is easily rearranged to give the **special eigenproblem for longitudinal motion**:

$$\begin{aligned} & \begin{bmatrix} R_{x,\mu} & R_{x,\alpha} & R_{x,\bar{q}} & 0 & 0 & -R_{gx} \cos \theta_o \\ \frac{R_{z,\mu}}{1 - R_{z,\hat{\alpha}}} & \frac{R_{z,\alpha}}{1 - R_{z,\hat{\alpha}}} & \frac{1 + R_{z,\bar{q}}}{1 - R_{z,\hat{\alpha}}} & 0 & 0 & -\frac{R_{gx} \sin \theta_o}{1 - R_{z,\hat{\alpha}}} \\ R_{m,\mu} + \frac{R_{m,\hat{\alpha}} R_{z,\mu}}{1 - R_{z,\hat{\alpha}}} & R_{m,\alpha} + \frac{R_{m,\hat{\alpha}} R_{z,\alpha}}{1 - R_{z,\hat{\alpha}}} & R_{m,\bar{q}} + \frac{R_{m,\hat{\alpha}} (1 + R_{z,\bar{q}})}{1 - R_{z,\hat{\alpha}}} & 0 & 0 & -\frac{R_{m,\hat{\alpha}} R_{gx} \sin \theta_o}{1 - R_{z,\hat{\alpha}}} \\ \cos \theta_o & \sin \theta_o & 0 & 0 & 0 & -\sin \theta_o \\ -\sin \theta_o & \cos \theta_o & 0 & 0 & 0 & -\cos \theta_o \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \\ & -\lambda \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} \chi_\mu \\ \chi_\alpha \\ \chi_{\bar{q}} \\ \chi_{\xi_x} \\ \chi_{\xi_z} \\ \chi_\theta \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{Bmatrix} \end{aligned} \quad (8.2.3)$$